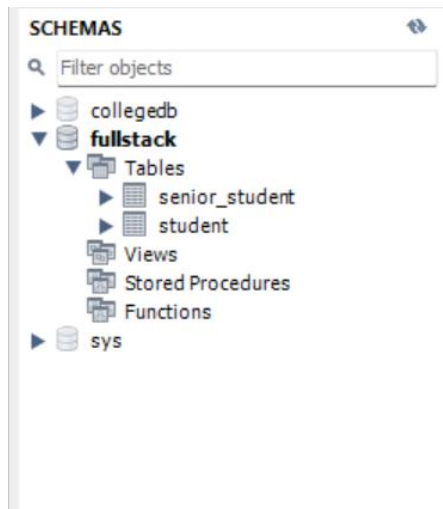


1. Create a database.

CREATE DATABASE fullstack;

USE fullstack;



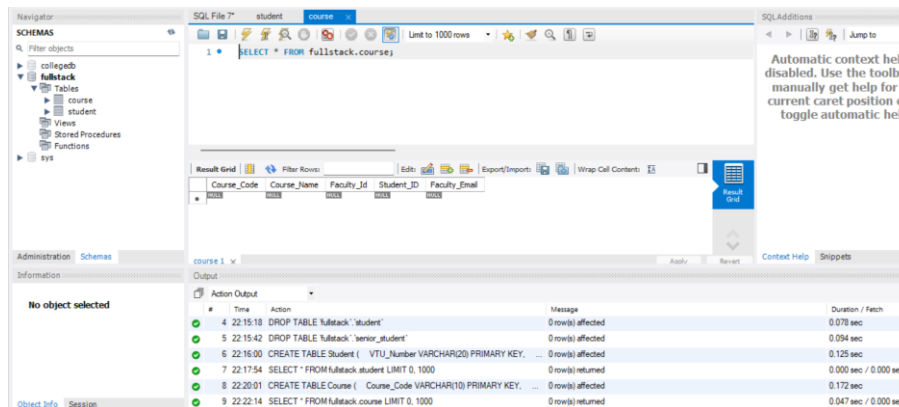
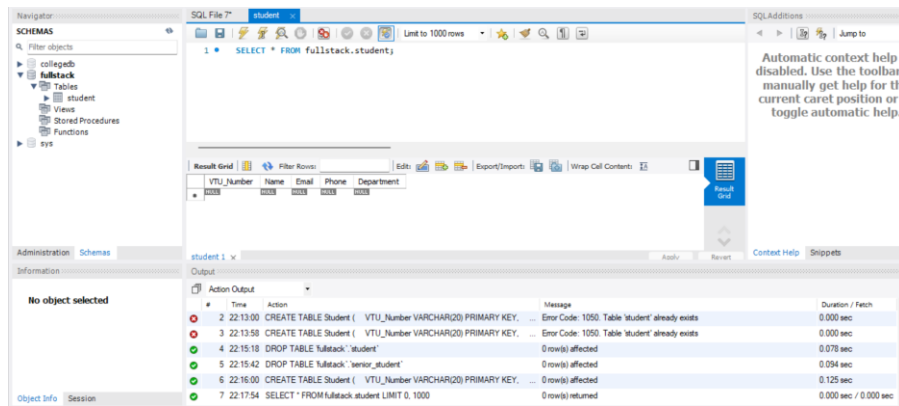
2. Create at least two tables:

Table 1: Student (VTU Number, Name, Email, Phone, Department)

```
CREATE TABLE Student (  
    VTU_Number VARCHAR(20) PRIMARY KEY,  
    Name VARCHAR(100),  
    Email VARCHAR(100),  
    Phone VARCHAR(15),  
    Department VARCHAR(50)  
);
```

Table 2: Course (Course Code, Course Name, Faculty Id, Student ID, Faculty Email)

```
CREATE TABLE Course (  
    Course_Code VARCHAR(10) PRIMARY KEY,  
    Course_Name VARCHAR(100),  
    Faculty_Id INT,  
    Student_ID VARCHAR(20),  
    Faculty_Email VARCHAR(100),  
    FOREIGN KEY (Student_ID) REFERENCES Student(VTU_Number)  
);
```



3. Insert minimum 5 records into each table.

Inserting into Student Table

INSERT INTO Student VALUES

('1MS23CS001', 'Arjun Kumar', 'arjun@mail.com', '9845011223', 'CSE'),
 ('1MS23CS002', 'Sneha Rao', 'sneha@mail.com', '9845022334', 'IT'),
 ('1MS23CS003', 'Rahul Nair', 'rahul@mail.com', '9845033445', 'CSE'),
 ('1MS23CS004', 'Priya Das', 'priya@mail.com', '9845044556', 'ECE'),
 ('1MS23CS005', 'Amit Singh', 'amit@mail.com', '9845055667', 'IT');

Inserting into Course Table

INSERT INTO Course VALUES

('CS101', 'Database Management', 101, '1MS23CS001', 'prof_smith@vtu.edu'),
 ('IS202', 'Data Structures', 102, '1MS23CS002', 'prof_jones@vtu.edu'),
 ('CS103', 'Operating Systems', 101, '1MS23CS003', 'prof_smith@vtu.edu'),
 ('EC301', 'Digital Electronics', 103, '1MS23CS004', 'prof_kumar@vtu.edu'),
 ('IS204', 'Web Programming', 104, '1MS23CS005', 'prof_leila@vtu.edu');

SQL File 7: student course

1 * SELECT * FROM fullstack.student;

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

VTU_Number	Name	Email	Phone	Department
IMS23CS002	Sneha Rao	sneha@gmail.com	9845022334	IT
IMS23CS003	Rahul Nair	rahul@gmail.com	9845033445	CSE
IMS23CS004	Priya Das	priya@gmail.com	9845044556	ECE
IMS23CS005	Amit Singh	amit@gmail.com	9845055667	IT

student 1 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	22:16:00	CREATE TABLE Student (VTU_Number VARCHAR(20) PRIMARY KEY, ...	0 row(s) affected	0.125 sec
7	22:17:54	SELECT * FROM fullstack.student LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
8	22:20:01	CREATE TABLE Course (Course_Code VARCHAR(10) PRIMARY KEY, ...	0 row(s) affected	0.172 sec
9	22:22:14	SELECT * FROM fullstack.course LIMIT 0, 1000	0 row(s) returned	0.047 sec / 0.000 sec
10	22:26:11	INSERT INTO Student VALUES ('IMS23CS001', 'Ajun Kumar', 'ajun@mail.c...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
11	22:29:50	SELECT * FROM fullstack.student LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

SQL File 7: student course student course

1 * SELECT * FROM fullstack.course;

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Course_Code	Course_Name	Faculty_Id	Student_ID	Faculty_Email
CS101	Database Management	101	IMS23CS001	prof_smith@vtu.edu
CS103	Operating Systems	101	IMS23CS003	prof_smith@vtu.edu
EC301	Digital Electronics	103	IMS23CS004	prof_kumar@vtu.edu
IS202	Data Structures	102	IMS23CS002	prof_jones@vtu.edu
IS204	Web Programming	104	IMS23CS005	prof_jeila@vtu.edu

course 1 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	22:20:01	CREATE TABLE Course (Course_Code VARCHAR(10) PRIMARY KEY, ...	0 row(s) affected	0.172 sec
9	22:22:14	SELECT * FROM fullstack.course LIMIT 0, 1000	0 row(s) returned	0.047 sec / 0.000 sec
10	22:26:11	INSERT INTO Student VALUES ('IMS23CS001', 'Ajun Kumar', 'ajun@mail.c...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
11	22:29:50	SELECT * FROM fullstack.student LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
12	22:32:35	INSERT INTO Course VALUES ('CS101', 'Database Management', 101, 'IMS...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.063 sec
13	22:32:41	SELECT * FROM fullstack.course LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

4. Select records using different CLAUSE.

WHERE clause:

SELECT * FROM Student WHERE Department = 'CSE';

SQL File 7: student course student course

24 ('CS101', 'Database Management', 101, 'IMS23CS001', 'prof_smith@vtu.edu'),
 25 ('IS202', 'Data Structures', 102, 'IMS23CS002', 'prof_jones@vtu.edu'),
 26 ('CS103', 'Operating Systems', 101, 'IMS23CS003', 'prof_smith@vtu.edu'),
 27 ('EC301', 'Digital Electronics', 103, 'IMS23CS004', 'prof_kumar@vtu.edu'),
 28 ('IS204', 'Web Programming', 104, 'IMS23CS005', 'prof_jeila@vtu.edu');
 29 * SELECT * FROM Student WHERE Department = 'CSE';

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

VTU_Number	Name	Email	Phone	Department
IMS23CS001	Ajun Kumar	ajun@gmail.com	9845011223	CSE
IMS23CS003	Rahul Nair	rahul@gmail.com	9845033445	CSE

student 1 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	22:22:14	SELECT * FROM fullstack.course LIMIT 0, 1000	0 row(s) returned	0.047 sec / 0.000 sec
10	22:26:11	INSERT INTO Student VALUES ('IMS23CS001', 'Ajun Kumar', 'ajun@mail.c...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
11	22:29:50	SELECT * FROM fullstack.student LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
12	22:32:35	INSERT INTO Course VALUES ('CS101', 'Database Management', 101, 'IMS...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.063 sec
13	22:32:41	SELECT * FROM fullstack.course LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
14	22:39:29	SELECT * FROM Student WHERE Department = 'CSE' LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

ORDER BY Clause (Sorting):

SELECT * FROM Course ORDER BY Course_Name ASC;

```

29 • SELECT * FROM Student WHERE Department = 'CSE';
30 • SELECT * FROM Course ORDER BY Course_Name ASC;
31

```

Course_Code	Course_Name	Faculty_Id	Student_ID	Faculty_Email
IS202	Data Structures	102	IMS23CS002	prof_jones@vtu.edu
CS101	Database Management	101	IMS23CS001	prof_smith@vtu.edu
EC301	Digital Electronics	103	IMS23CS004	prof_kumar@vtu.edu
CS103	Operating Systems	101	IMS23CS003	prof_smith@vtu.edu
IS204	Web Programming	104	IMS23CS005	prof_jella@vtu.edu

Course 2 x Apply Revert

Output

LIKE Clause (Pattern Matching):

SELECT Name, Email FROM Student WHERE Email LIKE '%@mail.com';

```

29 • SELECT * FROM Student WHERE Department = 'CSE';
30 • SELECT * FROM Course ORDER BY Course_Name ASC;
31 • SELECT Name, Email FROM Student WHERE Email LIKE '%@mail.com';
32

```

Name	Email
Arjun Kumar	arjun@mail.com
Sneha Rao	sneha@mail.com
Rahul Nair	rahul@mail.com
Priya Das	priya@mail.com
Amit Singh	amit@mail.com

Student 3 x Read Only

Output

Action Output

LIMIT Clause (Restricting Results):

SELECT * FROM Student LIMIT 3;

```

32 • SELECT * FROM Student LIMIT 3;
33

```

VTU_Number	Name	Email	Phone	Department
IMS23CS001	Arjun Kumar	arjun@mail.com	9845011223	CSE
IMS23CS002	Sneha Rao	sneha@mail.com	9845022334	IT
IMS23CS003	Rahul Nair	rahul@mail.com	9845033445	CSE
NULL	NULL	NULL	NULL	NULL

Student 4 x Apply Revert

Output

Action Output

#	Time	Action	Message
---	------	--------	---------

GROUP BY & HAVING Clause (Aggregating):

SELECT Department, COUNT(*) as Total_Students

FROM Student

GROUP BY Department

HAVING COUNT(*) > 1;

33 • `SELECT Department, COUNT(*) as Total_Students`
 34 `FROM Student`
 35 `GROUP BY Department`

Result Grid

Department	Total_Students
CSE	2
IT	2

Result 5 x Read Only

Session 3:

1. Write SELECT queries to display all records.

Display all students

SELECT * FROM Student;

36 `HAVING COUNT(*) > 1;`
 37 • `SELECT * FROM Student;`
 38

Result Grid

VTU_Number	Name	Email	Phone	Department
1MS23CS001	Arjun Kumar	arjun@mail.com	9845011223	CSE
1MS23CS002	Sneha Rao	sneha@mail.com	9845022334	IT
1MS23CS003	Rahul Nair	rahul@mail.com	9845033445	CSE
1MS23CS004	Priya Das	priya@mail.com	9845044556	ECE
1MS23CS005	Amit Singh	amit@mail.com	9845055667	IT

Student 6 x Apply Revert

Display all courses

SELECT * FROM Course;

37 • `SELECT * FROM Student;`
 38 • `SELECT * FROM Course;`
 39

Result Grid

Course_Code	Course_Name	Faculty_Id	Student_ID	Faculty_Email
CS101	Database Management	101	1MS23CS001	prof_smith@vtu.edu
CS103	Operating Systems	101	1MS23CS003	prof_smith@vtu.edu
EC301	Digital Electronics	103	1MS23CS004	prof_kumar@vtu.edu
IS202	Data Structures	102	1MS23CS002	prof_jones@vtu.edu
IS204	Web Programming	104	1MS23CS005	prof_leila@vtu.edu

Course 7 x Apply

2. Write queries using aggregate functions.

Count total number of students

SELECT COUNT(*) AS Total_Students FROM Student;

```

37 • SELECT * FROM Student;
38 • SELECT * FROM Course;
39 • SELECT COUNT(*) AS Total_Students FROM Student;
40

```

Result Grid

Total_Students
5

result 8

Count unique departments

SELECT COUNT(DISTINCT Department) AS Department_Count FROM Student;

```

39 • SELECT COUNT(*) AS Total_Students FROM Student;
40 • SELECT COUNT(DISTINCT Department) AS Department_Count FROM Student;
41

```

Result Grid

Department_Count
3

Find the highest Faculty ID assigned

SELECT MAX(Faculty_Id) AS Highest_Faculty_ID FROM Course;

```

40 • SELECT COUNT(DISTINCT Department) AS Department_Count FROM Student;
41 • SELECT MAX(Faculty_Id) AS Highest_Faculty_ID FROM Course;
42

```

Result Grid

Highest_Faculty_ID
104

result 10

- Sort data and display based in ascending/descending order of the VTU number.

Sort - Ascending order (Default)

**SELECT * FROM Student
ORDER BY VTU_Number ASC;**

```

40 • SELECT COUNT(DISTINCT Department) AS Department_Count FROM Student;
41 • SELECT MAX(Faculty_Id) AS Highest_Faculty_ID FROM Course;
42 • SELECT * FROM Student
43 ORDER BY VTU_Number ASC;

```

VTU_Number	Name	Email	Phone	Department
1MS23CS001	Arjun Kumar	arjun@mail.com	9845011223	CSE
1MS23CS002	Sneha Rao	sneha@mail.com	9845022334	IT
1MS23CS003	Rahul Nair	rahul@mail.com	9845033445	CSE
1MS23CS004	Priya Das	priya@mail.com	9845044556	ECE
1MS23CS005	Amit Singh	amit@mail.com	9845055667	IT

Sort - in Descending order

SELECT * FROM Student
ORDER BY VTU_Number DESC;

```

43 ORDER BY VTU_Number ASC;
44 • SELECT * FROM Student
45 ORDER BY VTU_Number DESC;

```

VTU_Number	Name	Email	Phone	Department
1MS23CS005	Amit Singh	amit@mail.com	9845055667	IT
1MS23CS004	Priya Das	priya@mail.com	9845044556	ECE
1MS23CS003	Rahul Nair	rahul@mail.com	9845033445	CSE
1MS23CS002	Sneha Rao	sneha@mail.com	9845022334	IT
1MS23CS001	Arjun Kumar	arjun@mail.com	9845011223	CSE

4. Display student records belonging to a particular department.

SELECT * FROM Student
WHERE Department = 'CSE';

```

45 ORDER BY VTU_Number DESC;
46 • SELECT * FROM Student
47 WHERE Department = 'CSE';

```

VTU_Number	Name	Email	Phone	Department
1MS23CS001	Arjun Kumar	arjun@mail.com	9845011223	CSE
1MS23CS003	Rahul Nair	rahul@mail.com	9845033445	CSE
*	NULL	NULL	NULL	NULL

Inner Join:

SELECT
s.VTU_Number,
s.Name AS Student_Name,

c.Course_Name,

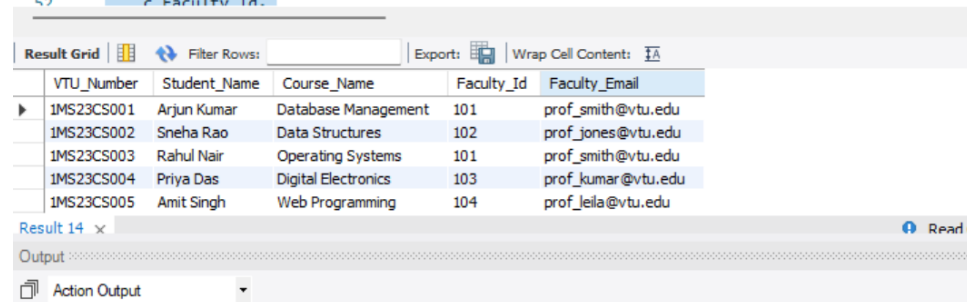
c.Faculty_Id,

c.Faculty_Email

FROM Student s

INNER JOIN Course c ON s.VTU_Number = c.Student_ID;

```
48 • SELECT
49     s.VTU_Number,
50     s.Name AS Student_Name,
51     c.Course_Name,
52     c.Faculty_Id,
```



The screenshot shows a database query result grid with the following data:

VTU_Number	Student_Name	Course_Name	Faculty_Id	Faculty_Email
1MS23CS001	Arjun Kumar	Database Management	101	prof_smith@vtu.edu
1MS23CS002	Sneha Rao	Data Structures	102	prof_jones@vtu.edu
1MS23CS003	Rahul Nair	Operating Systems	101	prof_smith@vtu.edu
1MS23CS004	Priya Das	Digital Electronics	103	prof_kumar@vtu.edu
1MS23CS005	Amit Singh	Web Programming	104	prof_leila@vtu.edu

Left Join:

SELECT

s.VTU_Number,

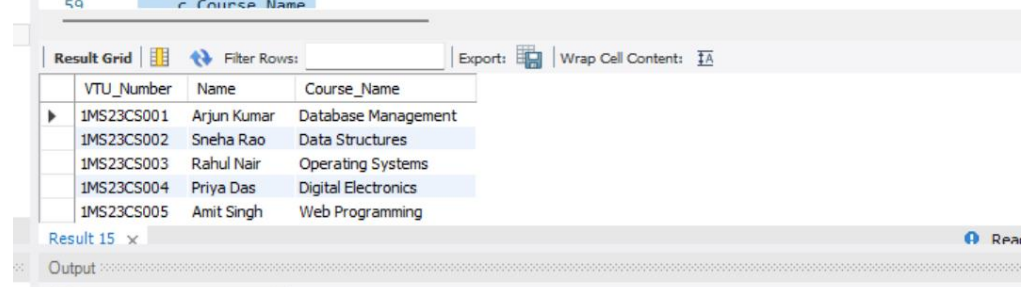
s.Name,

c.Course_Name

FROM Student s

LEFT JOIN Course c ON s.VTU_Number = c.Student_ID;

```
55     INNER JOIN Course c ON s.VTU_Number = c.Student_ID;
56 • SELECT
57     s.VTU_Number,
58     s.Name,
59     c.Course_Name
```



The screenshot shows a database query result grid with the following data:

VTU_Number	Name	Course_Name
1MS23CS001	Arjun Kumar	Database Management
1MS23CS002	Sneha Rao	Data Structures
1MS23CS003	Rahul Nair	Operating Systems
1MS23CS004	Priya Das	Digital Electronics
1MS23CS005	Amit Singh	Web Programming

Right Join:

SELECT

s.Name,

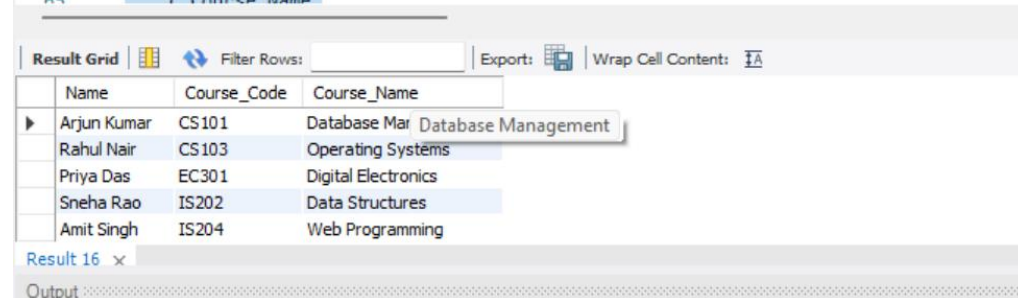
c.Course_Code,

c.Course_Name

FROM Student s

RIGHT JOIN Course c ON s.VTU_Number = c.Student_ID;

```
61 LEFT JOIN Course c ON s.VTU_Number = c.Student_ID;
62 SELECT
63     s.Name,
64     c.Course_Code,
65     c.Course_Name
```



Name	Course_Code	Course_Name
Arjun Kumar	CS101	Database Management
Rahul Nair	CS103	Operating Systems
Priya Das	EC301	Digital Electronics
Sneha Rao	IS202	Data Structures
Amit Singh	IS204	Web Programming

FULL JOIN (Full Outer Join):

SELECT s.Name, c.Course_Name FROM Student s

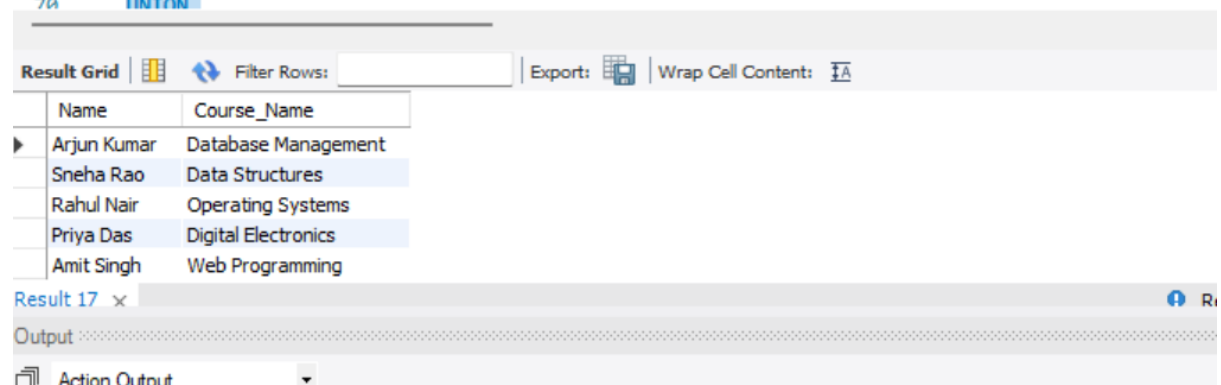
LEFT JOIN Course c ON s.VTU_Number = c.Student_ID

UNION

SELECT s.Name, c.Course_Name FROM Student s

RIGHT JOIN Course c ON s.VTU_Number = c.Student_ID;

```
67 RIGHT JOIN Course c ON s.VTU_Number = c.Student_ID;
68 SELECT s.Name, c.Course_Name FROM Student s
69 LEFT JOIN Course c ON s.VTU_Number = c.Student_ID
70 UNION
```



Name	Course_Name
Arjun Kumar	Database Management
Sneha Rao	Data Structures
Rahul Nair	Operating Systems
Priya Das	Digital Electronics
Amit Singh	Web Programming

Cross Join(Cartesian Product):

SELECT s.Name, c.Course_Name

FROM Student s

CROSS JOIN Course c;

```
72 RIGHT JOIN Course c ON s.VTU_Number = c.Student_ID;  
73 • SELECT s.Name, c.Course_Name  
74 FROM Student s  
75 CROSS JOIN Course c;
```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	Name	Course_Name
▶	Amit Singh	Database Management
	Priya Das	Database Management
	Rahul Nair	Database Management
	Sneha Rao	Database Management
	Arjun Kumar	Database Management

Result 18 ×

Output